

Climate drives variation in optimal phenology: 46 years of multi-environment trials in sunflower

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Received: 23 May 2025

Accepted: 20 October 2025

New Phytologist (2026) 249: 1527–1541

doi: 10.1111/nph.70790

Key words: agricultural field trials, flowering time, global change, *Helianthus annuus*, multi-environment trials (METs), phenology shift, selection, yield stability.

Summary

- Phenological shifts are near ubiquitous responses to climate change in both wild and agricultural systems, but the fitness consequences of these shifts are unclear. We evaluate how flowering phenology influences fitness and how climate influences the relationship between flowering phenology and fitness.
- We use a large dataset of performance trials of oilseed sunflower varieties (*Helianthus annuus* L.) conducted since 1978 across the Great Plains of North America. We estimate the flowering time that optimizes yield (fitness) in 341 environments to quantify how climate variation impacts the optimal flowering time.
- We find that temperature is a key driver of optimal phenology, with earlier and faster flowering favored in hotter years and locations. Flowering time differences explain 9% of the variation in fitness between sunflower genotypes within trials. Flowering shifts a week away from the optimum incurred substantial penalties, with a median yield reduction of 23%.
- Our results indicate that the shifts to earlier phenology commonly observed under climate warming may be beneficial. In sunflower, maximizing agricultural productivity under future climates will likely require both careful selection of varieties to match new conditions as well as breeding new varieties that flower faster than material that is currently available commercially.

Introduction

Shifts in phenology, or the timing of recurring seasonal events such as spring emergence, migration, reproduction, and dormancy, are some of the most frequently observed biological changes in response to climate change (Parmesan & Yohe, 2003; Parmesan, 2006; Lane *et al.*, 2012; Menzel *et al.*, 2020). Both plastic (Zettlemoyer & Peterson, 2021; Knott *et al.*, 2023; Vahsen *et al.*, 2025) and evolutionary (Bradshaw & Holzapfel, 2001; Franks *et al.*, 2007) responses to climate change have been observed in phenological traits. However, despite the near-ubiquity of phenology shifts in the face of climate change, it is often unclear to what degree these changes in timing confer beneficial – or detrimental – effects on fitness. In both natural and agricultural environments, a better understanding of the impacts of phenology on organismal performance in the face of climate variation is essential for understanding biological systems under environmental change, as well as designing successful conservation, restoration, and food security programs.

Optimal flowering phenology depends on a complex set of factors, which must balance the benefits and risks of flowering at a particular time (Wolkovich *et al.*, 2014). Shifts to earlier

flowering phenology could benefit plants if they avoid stressful conditions during hotter seasons, while delaying flowering may be beneficial as it leaves more time for growth, development, and improving nutritional status before reproductive investment (Johansson *et al.*, 2013; Kurze *et al.*, 2021). Timing can also affect biotic interactions, particularly if species differ in their responses to climate (Lang *et al.*, 2025), with the ideal timing facilitating beneficial interactions while avoiding predation and competition (Park & Post, 2022). The complexity of selection pressures on flowering timing means that the optimal timing is typically difficult to predict. Few studies have examined the degree to which observed phenological shifts are in the same direction as selection, and what the optimal phenology should be.

The studies that do examine selection on phenology show that earlier spring phenology is often favored. In a meta-analysis, spring phenological events of birds were consistently selected to be earlier across decades, resulting in adaptive change in a majority of studies (Radchuk *et al.*, 2019). Austen & Weis (2015) found that for non-domesticated *Brassica rapa*, individuals with an earlier flowering date produced more seed regardless of age or size of the plant, indicating the temporal environment

predominantly determines optimal flowering time, rather than the condition or age of the plant. A meta-analysis of plants also found that, on average, selection favored earlier flowering and that selection for earlier flowering was strongest in studies that occurred in temperate environments (Munguía-Rosas *et al.*, 2011). However, in this meta-analysis, the species that occurred in temperate sites differed from those in tropical sites, making it difficult to disentangle whether selection gradients for individual species differed among locations. **Models suggest that selection gradients for phenological timing vary over space and time** (Pau *et al.*, 2011; Gauzere *et al.*, 2020). Yet empirical work quantifying how optimal phenology is driven by changes in both spatial and temporal variation in climate, holding species constant, is largely lacking.

It is urgently necessary that agricultural systems continue to adapt to climate change to keep up with the increasing demand for food and feed (Howden *et al.*, 2007). Phenology has long been known as a major determinant of crops' ability to thrive in new environments (Rathcke & Lacey, 1985), and there is a long history of selection to modify the phenology of wild progenitors as part of crop domestication (Evans, 1980). **Phenological responses to climate change have already been observed in some agricultural systems** (Xiao *et al.*, 2013; Tariq *et al.*, 2018; Menzel *et al.*, 2020). Menzel *et al.* (2020) found that planting and harvest have advanced and the growing season has lengthened, but that the period between flowering and fruit maturity has shortened for crops in three central European countries. Overall, however, **climate-induced changes in phenology have been relatively under-studied in agricultural systems compared to wild populations** (Hassan *et al.*, 2024), despite the obvious importance of ensuring stability in food supplies. Because **phenological shifts can differ between wild and cultivated populations** (Bartošová *et al.*, 2025), we cannot assume that crop and nearby wild populations will behave in the same way. Moreover, it remains unresolved whether phenological shifts in crops are adaptive to climate change, for example by maintaining or increasing yield.

Quantifying the fitness landscape – and how it will change with climate shifts – requires both phenology and fitness measurements on multiple individuals, replicated with standardized treatment conditions in many sites with different climates (common gardens). In wild systems, these data are extremely rare. In contrast, agricultural field trials are a rich but underused resource. In multi-environment agricultural trials, the same genotypes of a crop are grown in multiple locations each year as a means of testing the climatic suitability of each genotype and to guide variety selection for farmers. Genotypes are often replanted for multiple years until they are replaced with newly developed genotypes. These multi-environment longitudinal trials, in which both phenology and yield are measured, can be used to understand both how phenology contributes to yield and how optimal phenology varies across location and climate.

Here, we use 46 years of multi-environment field trials of commercial cultivars of sunflower (*Helianthus annuus* L.) in the Great Plains region of the United States, encompassing 37 sites and 2158 unique genotypes, to examine the relationships between flowering phenology, yield, and climate. Yield of annual

sunflowers is the total reproductive output of a sunflower genotype and is the agricultural equivalent of fitness in wildland species. We acknowledge that this is an agricultural setting and most seeds will not be replanted, but yield is a performance metric and a good proxy for fitness, as both quantify the reproductive output of sunflower genotypes. We ask specifically, (1) How important is phenology in explaining variation in performance? (2) How often is a commercial variety available that flowers at the optimal time? (3) How do geographic and interannual differences in weather influence optimal flowering time? and (4) **How will global warming impact optimal flowering time?**

Materials and Methods

Sunflower natural history

Sunflower (*Helianthus annuus* L.) is an annual flowering plant with a widespread geographic distribution. Sunflower has a native range across much of North America and has been introduced to all continents other than Antarctica. The same species that occurs in wild and naturalized populations also has been domesticated as an economically important food plant, particularly as a source of oil, as well as for ornamental and other purposes. The main growing regions of sunflower, globally, have similar climates, with the top producing countries being in Russia, Ukraine, the European Union, and Argentina (USDA Foreign Agricultural Service, 2025). Sunflower production in the United States is concentrated in the Great Plains region, with North Dakota and South Dakota planting the largest land area of sunflower (Sandbakken, 2023). The climate in which sunflower is grown impacts the grain yield and oil content of the seeds. Sunflower growth and development can occur between 6.7°C (Kandel *et al.*, 2020) and *c.* 26–29°C, with an optimal daytime temperature of 25°C (Kalyar *et al.*, 2014). Modern varieties of sunflower generally produce a single inflorescence, and senescence begins following seed set. Thus, as in other semelparous species, the onset of reproduction is closely tied to the many developmental and metabolic changes associated with not only reproduction, but also senescence. The genetics of the flowering time pathway has been well studied in sunflower (Blackman *et al.*, 2011a,b; Cadic *et al.*, 2013; Badouin *et al.*, 2017), making it an excellent system for understanding the fitness consequences of phenology and its relationship with climate.

Several factors influence optimal flowering time for sunflowers. Seed loss from blackbirds and seed-infesting insects, such as the red sunflower seed weevil, increases later in the season (Oseto *et al.*, 1987; Clark *et al.*, 2020). However, **increasing the length of the vegetative and reproductive stages by delaying flowering may be useful for biomass accumulation** (Barros *et al.*, 2004). During the grain-filling stage (starting *c.* 15 d after flowering), sunflowers are more sensitive to weather, and heat or cold stress during this time can reduce both yield and oil content, and also alter the types of fatty acids in the oil (Kalyar *et al.*, 2014; van der Merwe *et al.*, 2015). Thus, there are multiple factors influencing optimal flowering time and it is not clear whether earlier or later flowering strategies improve yield, or how interannual variation

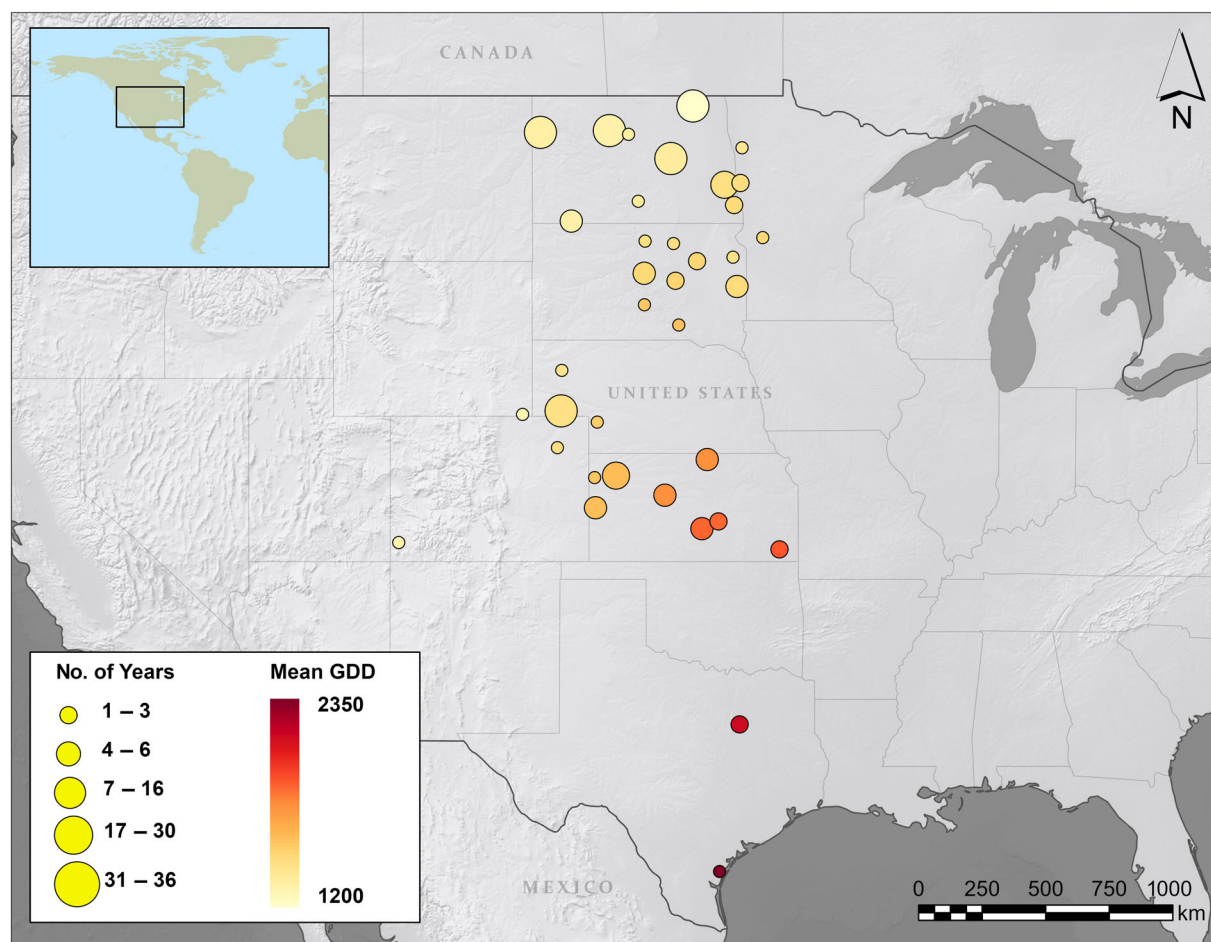


Fig. 1 Sites of sunflower multi-environment trials. The size of the dot represents the number of years of data at each location, and the color shows the average growing degree day (GDD) accumulation of 125 d during a typical growing season (May–August). Map made in ESRI ArcGIS Pro (Redlands, CA). Basemap data source: Natural Earth (public domain). Map lines delineate study areas and do not necessarily depict accepted national boundaries.

in climate influences optimal timing. Genotypes differ in their responses to heat stress (van der Merwe *et al.*, 2015), and in the temperature and daylength requirements for flowering; thus, selecting varieties for cultivation that are well matched with the climate is important to yield.

Phenology and performance data

To determine how flowering time impacts performance over space and time, we used data from commercial sunflower variety performance trials, also known as multi-environment trials. In performance trials, commercial varieties (genotypes) are planted at one or more field sites. Trials are intended to mimic the planting dates, growing conditions, and management practices of commercially grown sunflowers, and trial sites are often adjacent to commercial fields. Genotypes are generally chosen to be included in a field trial if companies believe the genotype will perform well in that environment. Thus, the genotypes in the trials already represent elite genotypes that have been selected for the locations and do not represent all genetic variation available in the crop.

Data from sunflower field trials were compiled from published records of each trial, digitized, and checked for quality (S. C. Elmendorf *et al.*, in prep). The dataset contains trial-level means for each genotype for a variety of performance and morphological traits. Here, we focus on non-irrigated trials of oilseed sunflower varieties that measured both yield and flowering phenology. Yield is the pounds of sunflower seeds (corrected to 10% moisture) per acre and represents the total reproductive output (fitness surrogate) of a genotype. Flowering phenology is the day on which it was observed that 50% of the plants of a genotype flowered (stage R5.1; Schneiter & Miller, 1981). Because sunflowers have showy flowers and tend to flower synchronously within genotypes, this is a reliable measure of flowering phenology that can be consistently scored by different observers across trials and years. The planting density of sunflower does not generally have a large effect on seed yield (Barros *et al.*, 2004; Olson *et al.*, 2024) and many records did not include this information, so we did not correct for potentially differing planting densities across trials.

We analyzed data on yield and phenology from 37 locations from North Dakota to Texas (Fig. 1), and 46 years, for a total of 341 site-years. Each location had trials for 1 to 36 years (mean

9 years per location). Between 4 and 137 (average 43) genotypes (hybrid sunflower varieties) were planted per trial, with a total of 2158 unique genotypes represented. Within a trial, there is generally substantial variation between genotypes in phenology and yield. Data for each genotype is a trial-level average over all individual plants. Usually, trials were repeated at the same location (generally a research station) for all years; however, some trials moved from year to year between several nearby locations within the same county. We considered trials from the same county as single sites, in order to include more trials with repeated locations. Rarely, multiple trials were planted in the same county in a single year. After excluding trials that specifically manipulated planting date (e.g. re-crop plantings), trials in the same county in the same year were combined if the planting dates differed by no > 2 d; but the larger trial was chosen if planting dates differed by > 2 d, which excluded two trials. Fourteen trials in which all genotypes flowered within 5 d were excluded, due to concerns that phenology may not have been rigorously measured in those trials and that we would lack sufficient power to detect phenology–yield relationships.

Climate data

Daily climate summaries were extracted from the Daymet database (Thornton *et al.*, 2022) for all locations and all available years of data. The Daymet dataset is a 1 km $\times$ 1 km gridded dataset interpolated from weather station data, with records starting in 1980 (Thornton *et al.*, 2022).

To better describe the climatic conditions experienced by sunflowers, we calculated climate statistics for all years of available climate data. We focused on temperature, precipitation, and frost timing. To capture the temperature experienced by genotypes in their growth stage, we calculated the number of cumulative growing degree days for the 70-d period starting on the planting date of the trial with lower and upper thresholds of 6.7°C and 26°C, respectively (flowering-GDD) (Kalyar *et al.*, 2014; Kandel *et al.*, 2020). Seventy days represents the average time to flowering across the entire dataset, and the temperature thresholds were chosen based on established cold and heat stress that occur for sunflowers. For years in which there was no trial planted at a location, or the planting date was not recorded, we used the average planting date at that site. Flowering-GDD depends on both planting date and site temperature. Because climate may influence planting date, flowering-GDD calculations may reflect management responses to climate, rather than purely climate variability. To ensure our results are robust to the GDD window and also disentangle the effects of temperature of a site from management decisions, we calculated a site heat index as the cumulative growing degree days for a fixed period from day of year 125 to 250 at each location (site-GDD). This 125-d window was chosen to encompass planting through flowering of nearly all trials in the dataset. Precipitation in millimeters was totaled from 21 d before to 30 d after the average flowering date at each site, during which period yield is most susceptible to water stress (Stegman & Lemert, 1981). The frost day was the first day after flowering when the minimum daily temperature was below

-3.9°C (Berglund, 2004), a killing frost for a sunflower before maturity, or 366 if there was no day with a minimum temperature below -3.9°C .

Across the sunflower-growing region, there is a strong latitudinal gradient in temperature (Fig. 1). In the period from 1980 to 2023, most sites have warmed and the day of first frost has moved later, while trends in precipitation have been largely stable (Supporting Information Fig. S1).

Optimal flowering time estimation

To determine the relationship between flowering phenology and performance, we calculated the flowering date that was associated with the highest yield, which we call the ‘optimal flowering time’, for each trial. Each trial consists of data from an average of 43 genotypes. We fit a quadratic model with Gaussian error and log-link function, in which the flowering day for each genotype (scaled within site) and the square of flowering day predicted its yield. We excluded 10 counties in which only one trial had been performed. We fit a model for each site (27 total) and fit a separate intercept for each year within site. This method is equivalent to fitting each site-year separately (and also equivalent to fitting all sites and years in a single model), except that there is a single error term for each site rather than each site-year (Edwards & Crone, 2021). Thus, we estimated three coefficients per site-year, for flowering day (β_1), the square of flowering day (β_2), and an intercept for each year (α). We also repeated the model fitting, grouping trials that were planted in the same precise location (usually a research station) and excluding sites that did not have at least two trials, to make sure the grouping did not influence the results. We estimated the peak of a Gaussian curve (optimal flowering time) to be $\beta_2/(-2\beta_1)$. The 95% confidence interval and 1 SD interval were obtained through bootstrapping, following methods and code developed by Edwards & Crone (2021).

When the quadratic model yields a concave-up curve (if either β_1 was negative or β_2 was positive), an optimal flowering day cannot be estimated from a quadratic model, though there may be a strong relationship between flowering day and yield. For some concave-up quadratic models, we noted that the relationship between flowering timing and yield was essentially linear, with an optimum outside the range of the data. Therefore, rather than discard all site-years with concave-up curves (where an optimal flowering day could not be estimated), we also fit an alternative linear model, with log-link as above, to all trials. We then compared the corresponding quadratic and linear models. Where the slope of the fitted linear model was significantly ($P < 0.05$) different from zero and an AIC-based model comparison between the fit of the linear and original quadratic model did not indicate substantially better fit of the quadratic model ($\Delta\text{AIC} < 2$), and the optimal flowering time fell outside of the range of the data (either earlier or later than any variety planted), we used the linear, rather than quadratic model, to bound the estimated optimum flowering time (see treatment of censored records in question 3, subsequently). AIC values were calculated from models fit on individual site-years.

For each site-year ($n = 341$), we then calculated the optimal flowering time and its uncertainty as follows. Where the

quadratic model was used, the optimal flowering day was estimated from the peak of the fitted relationship between flowering phenology and yield as $\beta_1/(-2\beta_2)$, and SD of the peak as $\sqrt{1/(-2\beta_2)}$ (Edwards & Crone, 2021). We used parametric bootstrapping with 10 000 samples to calculate the 95% confidence intervals around the peak of the curve, as well as 1 SD from the peak (68% confidence interval). Where the linear model was used, we estimated that the optimal flowering time occurred before (negative slope) or after (positive slope) the earliest (latest) observed flowering date. Site-years in which the quadratic model was concave up, but the quadratic model was preferred by $\Delta\text{AIC} > 2$, or the slope of the linear model was not significant, did not have a clear optimal flowering time.

Importance of phenology (Question 1)

We used two methods to determine the importance of phenology: a meta-analysis on R^2 and the effect on yield of flowering 1 wk from optimal flowering. We analyzed through meta-analysis a generalized form of the coefficient of determination (R^2). We used raw McFadden's pseudo- R^2 $\left(1 - \frac{(\text{Deviance})}{(\text{Null deviance})}\right)$ from the generalized linear models as the effect size in the meta-analysis. This statistic is a ratio indicating how well the model fits. A high R^2 indicates that flowering time is a good predictor of yield within the trial, while a low R^2 indicates that flowering time is not a good predictor of yield.

We calculated R^2 for each trial by fitting a quadratic model, as above, but with a separate model for each trial, rather than all years of one site in a single model as above (the coefficients estimated by either approach are the same; however, R^2 can only be estimated for each trial if each trial is in a separate model). We used exclusively quadratic models in these analyses to avoid comparing models with different numbers of terms. The sampling variance was calculated with the large-sample approximation given by Kendall & Stuart (1979; Viechtbauer, 2010). We performed multilevel linear (mixed-effects) meta-regressions on the R^2 effect sizes with the METAFOR package (Viechtbauer, 2010) in order to appropriately propagate uncertainty. R^2 often increases non-linearly at low sample sizes (e.g. in the most extreme case, a linear regression of two points will have an R^2 of 1), and this was the case in our data (Fig. S2). To account for this bias, we included the sample size and (sample size)² as covariates in all models. To calculate the overall amount of variation explained, we estimated the global average R^2 at average sample sizes in a model with no additional covariates. We also conducted an additional analysis to test for temporal and regional differences in the importance of flowering time to yield, including both year and state as covariates in the model. In all models, we fit random intercepts for counties. We used the ORCHARD package for plotting (Nakagawa *et al.*, 2023). Data from 341 trials were included in these analyses.

As a second analysis of the importance of phenology to yield, we calculated the predicted change in yield for each trial when flowering was offset 7 d from the optimal flowering date. For trials in which the optimal flowering time was estimated to be

within the range of observed flowering dates (quadratic models), the estimated yield 7 d after the optimal flowering date was subtracted from the predicted yield at the optimal flowering date and divided by the yield at optimal flowering to estimate the change in yield (on a percentage basis) with change in flowering time. For trials in which the optimum was estimated to be outside the data range and the linear model had a positive slope, the predicted yield 7 d before the last day of flowering was subtracted from the predicted yield at the last day of flowering and scaled by the predicted yield at the last day of flowering. For linear models with a negative slope, the predicted yield 7 d after the first day of flowering was subtracted from the predicted yield on the first day of flowering and scaled by the predicted yield on the first day of flowering. We excluded trials in which the relationship between yield and flowering day was unclear (21% of the dataset, see next section). We used a bootstrap procedure with 10 000 iterations to estimate the 95% confidence intervals around the median decrease in yield across the dataset. Data from 263 trials were included in this analysis.

Availability of commercial varieties at the optimal flowering time (Question 2)

We next determined the suitability of available genetic material (genotypes) to trial conditions based on the date of the estimated optimal flowering time relative to the observed flowering times of genotypes planted. We categorized all trials into four categories, as follows. If the optimal flowering time was estimated to be within the range of observed flowering dates, the set of varieties was considered suitable for the environment (Fig. S3A). If the optimal flowering time was estimated from the quadratic model to be before or after all genotypes flowered in a trial, trials were categorized as flowering too late or too early, respectively, and mismatched with the environment. All trials for which the linear model was chosen were categorized as flowering too early (positive slope) or too late (negative slope) (Fig. S3B,C). Trials in which the quadratic model was concave up and AIC indicated a better fit of the quadratic than the linear model or the linear model was not significant were categorized as having an unclear optimal flowering time, or no apparent relationship between flowering time and yield (Fig. S3D).

We then determined how the categorization of each trial affected the average absolute yield in that trial. While the categorization of trials into suitable, early, and late was based on relative yield within a single trial, we sought to determine the impact on absolute yield, relative to all other trials. We included year and county as random effects, since some locations and years perform better than others in general.

Next, we determined whether the categorization of each trial was driven by temporal or climatic factors by fitting two multinomial regressions, with the four categories as the unordered dependent variable. To determine the temporal effect, we fit a model with year (continuous) as the only predictor. To determine the effect of climate, we fit a model with five climate variables. To separate the effect of climate across location and year, we calculated a site mean for flowering-GDD, precipitation, and frost

day, and the anomaly from the site mean for each year. Growing degree day and frost date site means were strongly correlated with each other (sites with more growing degree days also had later first frost dates), so frost date site mean was dropped from the model. Thus, site mean and anomalies for growing degree days and precipitation, and anomalies for frost date were all used as predictors in the model. For both temporal and climatic models, random intercepts were included for location of trial (county). Models were fit in brms (Bürkner, 2017). For the climate model, we focused on the climate predictors for which confidence intervals for the coefficients, which reflect the probability of a trial being too early (or late) vs having suitable timing, did not overlap zero. Predictions of marginal effects were made by holding all other factors in the model at their means. Data from 325 trials were included in the temporal and climate analyses.

Optimal flowering time across climate (Question 3)

Next, we sought to determine how variation in the optimal flowering time is driven by climate conditions and agronomic practices. We fit a Bayesian model that predicted how optimal flowering time changes with flowering-GDD, precipitation, day of first frost, and planting date, using the site means and anomalies as described above. As above, the frost date site mean was dropped from the model due to a strong correlation with growing degree day site means. The planting date of each trial was included as a covariate, and the location of each trial (county) was included as a random intercept.

Our variable (optimal flowering time) consisted of some site-years where optimal flowering was estimated – but with measurement error – as well as some site-years where optimal flowering time was known only to be before or after some value (censored records). We therefore used a combined measurement error and censored modeling technique to test the relationship between climate variables and optimal flowering in a single model that accommodated both types of data. We used brms (Bürkner, 2017) to generate separate initial models for each type of data using a non-centered parameterization to facilitate convergence, and then combined the resulting stan code to run a single model. (See Methods S1 for details.)

To ensure our results were robust, we fit several alternative models. Planting date influences the growing degree days accumulated, since later planting will subject the plants to hotter growing conditions. Because the interval for growing degree days started on the planting date, this could have potentially caused a spurious correlation between planting date and growing degree days. To test this and make sure our results were robust to our choice of interval for growing degree days, we reran the model with the fixed period growing degree days (from days 125 to 250, site-GDD), which represents only the climate of a site and not agronomic practices at that site. We also ran separate models for the optimal flowering day of the year and for optimal flowering days past planting. We also ran separate models grouping sites by county and by exact location. Data from 255 trials were included in these analyses.

We used the fitted model to predict the optimal flowering day across the range of site mean growing degree days and planting dates that were represented in the data. We made these predictions at the 5th, 50th, and 95th percentile growing degree day anomalies, representing a very cold year, an average year, and a very hot year. In these predictions, precipitation site mean and anomaly, and frost date anomaly were set at their means.

Optimal flowering time in the future (Question 4)

Finally, we sought to determine how optimal flowering will change in a warmer future. We used climate projections from the Multivariate Adaptive Constructed Analogs (MACA) (Abatzoglou & Brown, 2012) dataset (accessed at <https://climate.northwestknowledge.net/MACA/index.php>), which downscales 20 global climate models to *c.* 5 km scales. We used the 'Boxplot of Climate Projections' tool and selected on the map the region with the most sunflower agriculture (North and South Dakota, Nebraska, and Kansas) to estimate the multimodel ensemble projected increases in daily minimum and maximum temperature for the summer period (June, July, and August) in mid-century (2040–2069) for an intermediate emissions (RCP 4.5) scenario. We then added the projected increase in minimum (2.44°C) and maximum (2.89°C) daily temperature to the historical daily Daymet data for all years (1980–2023). We then calculated the growing degree day accumulation (flowering-GDD), as above, for the 70-d period starting at the planting date for each site to estimate typical temperatures experienced by sunflowers under the future climate scenario. At these projected temperatures, the average mid-century year would accumulate 109 more flowering-GDDs than the current average year. We predicted the optimal flowering time across the range of current site mean flowering-GDDs and planting dates, as above, except with a growing degree day anomaly of 109. 255 trials were included in this analysis.

Results

How important is phenology?

We found that the pseudo- R^2 was 8.9% (95% CI 5.14, 12.64) on average, after correcting for the effect of sample size of each trial, indicating that flowering timing is a meaningful predictor of yield within each trial. Over the 46 years of the study, R^2 has decreased, by 1.35 percentage points per decade (Fig. S4). There is both variation between and within sites in R^2 and, in particular, flowering time is a better predictor of yield in Texas, with an R^2 over 50% which is the hottest and most southern state in our dataset (Fig. 2a).

The yield penalty for flowering 1 wk before or after the optimal flowering date varied between 0 and 100% across the trials in our dataset, with a median yield decrease of 20% (95% CI 17.12, 22.27) compared to the yield at optimal flowering (Fig. 2b). The yield penalty did not statistically clearly differ between trials fitted with quadratic and linear models.

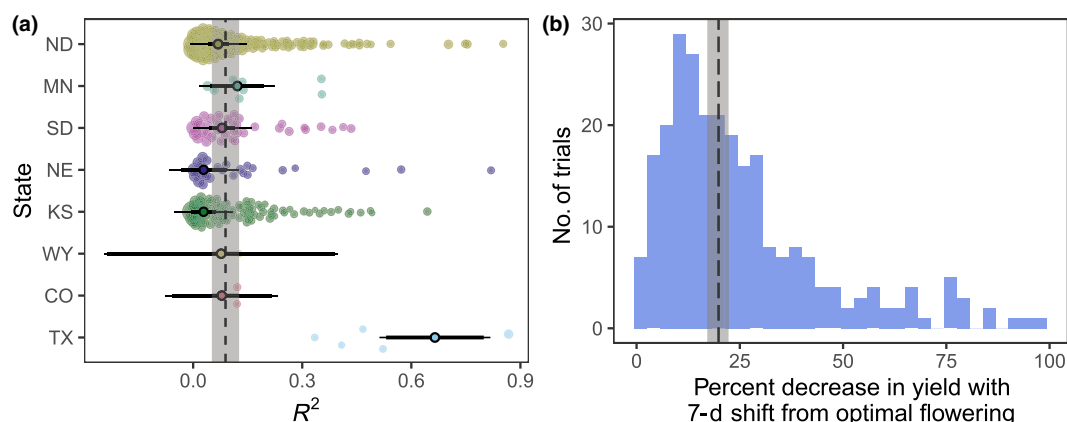


Fig. 2 Importance of phenology in sunflower. (a) R^2 from Gaussian models of flowering time and yield at each site-year, grouped by state. States are ordered roughly from north to south. Thick lines represent 95% confidence intervals, thin lines represent prediction intervals. Size of colored points represents the precision of the estimate, which is the inverse of the SE. Dashed vertical line and gray shaded area show the average R^2 across all states and 95% confidence interval. With sufficiently large sample sizes, the confidence interval will always be greater than zero; however, the intention of the analysis was to determine the average effect size, not whether the effect is above zero. (b) Percent decrease in yield with a 7-d shift in flowering time of trials in which there was a quadratic or linear relationship between flowering day and yield (80% of the entire dataset). Dashed vertical line and gray shaded area show the median decrease in yield across the trials and bootstrap 95% confidence interval around the median.

How often is a commercial variety available that flowers at the optimal time?

We found that 55% of trials included suitable genetic material for the environmental conditions (i.e. the optimal flowering time was estimated to be within the range of the observed flowering times). All genotypes in a trial flowered earlier than optimal in 16% of trials and later than optimal in 8% of trials. In 21% of trials, it was unclear whether there was an optimal flowering time in that trial from the data available. Trials with suitable genotypes tended to become more common over time, but this relationship was not statistically clear (Fig. S5).

Average yield was 14% lower in trials where all genotypes flowered too late compared to trials in which genotypes appeared to be suitable. However, there was no yield penalty for flowering too early, and a much smaller (4%) yield penalty for trials in which the relationship between flowering time and yield was unclear (Fig. 3).

Average site temperature and the timing of frost both influenced the odds that a trial included suitable genotypes, while temperature anomaly, precipitation site mean, and precipitation anomaly were not clear drivers. At warmer sites, the probability of a phenological match decreased, along with the probability of flowering too late, while unclear relationships became more frequent (Fig. 4a). In years with early first frost, trials were more likely to flower later than optimal, while years with a late frost were more likely to flower at the optimal time or have unclear optima (Fig. 4b).

How does climate influence the optimal flowering time?

There was no single optimal flowering time across the sites in our study. Rather, optimal flowering time varied over sites and across years within sites, ranging from 50 to 108 d after planting (CV = 0.13; Fig. S6).

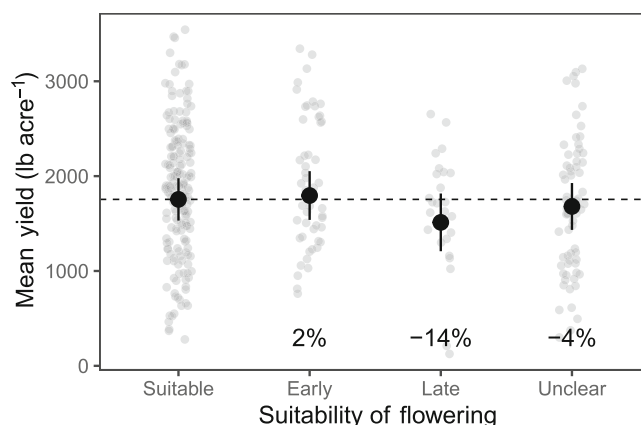


Fig. 3 The influence of flowering time on absolute yield in sunflower. Points and lines represent the average yield of trials in each category and 95% confidence intervals. Percentages show the change in yield relative to the 'Suitable' category. For comparison, the horizontal dashed line indicates the mean yield in the 'Suitable' category.

We found that temperature, measured as growing degree days in the 70 d following planting, was strongly associated with the optimal number of days to flowering, such that more rapid flowering was favorable in warm sites and years, and slower flowering in colder sites and years (Fig. 5a,b; Table 1). Our results imply that the optimal flowering time is 3 d earlier for every 100 degree days warmer a site is (over space). Similarly, optimal flowering is 5 d earlier in a year that is 100 degree days warmer than average. Planting day of year also influences the optimal flowering times (Fig. 5c; Tables 1, S1), such that when planting is later, more rapid flowering is favored.

We found no statistically clear relationship between optimal days to flowering and either precipitation (Fig. 5d,e) or first frost anomaly (Fig. 5f; Table 1).

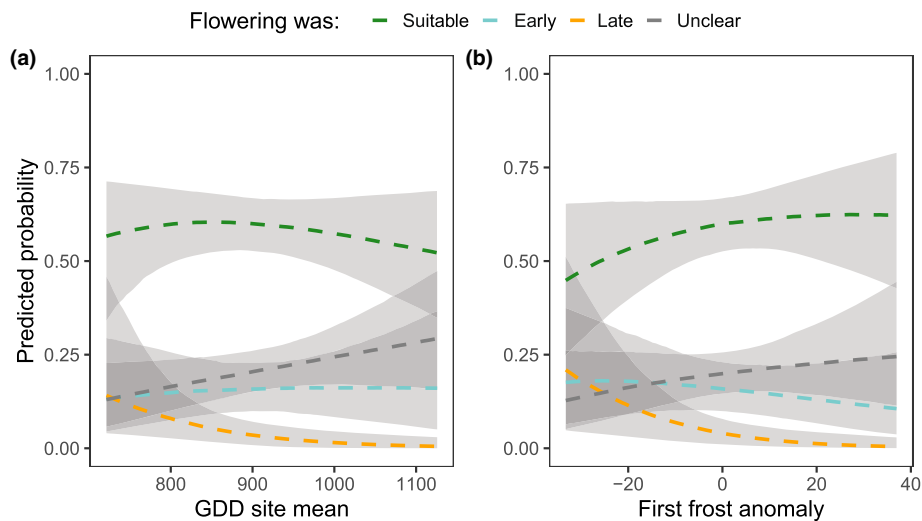


Fig. 4 Predicted probabilities that a trial will include genotypes that flower at the optimal flowering time, based on multinomial regression. The unclear category indicates trials that did not have a clear relationship between phenology and yield. (a) Predicted probabilities across growing degree day (GDD) site means, which range from cooler to warmer sites during a 70-d period after the planting date. (b) Predicted probabilities across first frost anomalies (units in days). Negative frost anomalies indicate years for which the first frost is earlier than average for a site, and positive values indicate a later first frost than average for a site. Shaded areas are 95% confidence intervals.

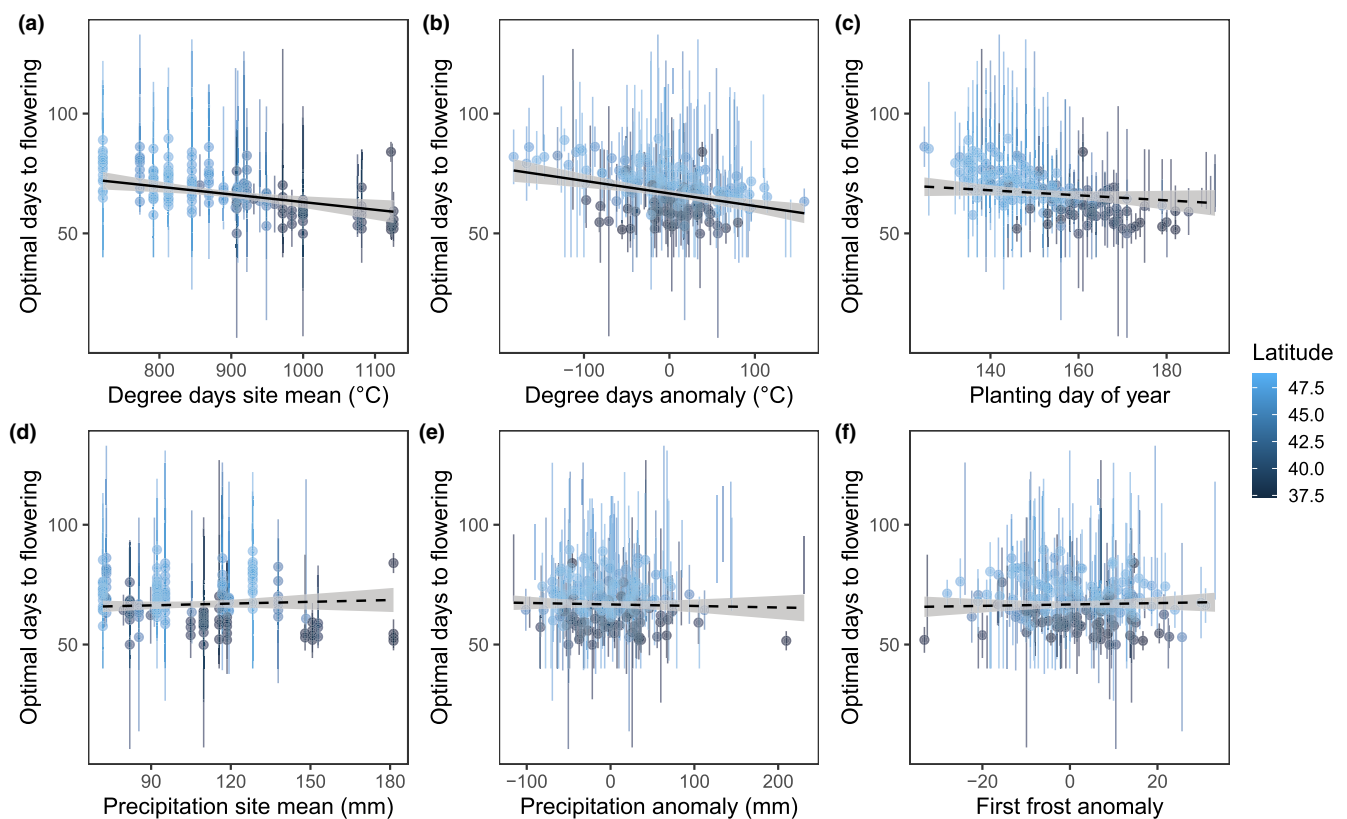


Fig. 5 The relationships between degree days (a, b), precipitation (d, e), frost timing (f) and planting day of year (c) and optimal days to flowering in sunflower. Black trend lines represent marginal effects over the range of one variable, with all other variables held constant at their mean, with solid lines indicating statistically clear effects. The shaded area is the 95% credible interval. Points are raw data for optimal days to flowering and measurement error. Vertical lines without points represent the censoring interval (see 'Materials and Methods' section) for trials with an optimal flowering time outside the range of data. Points and vertical lines are colored by the latitude of the site. The raw data points are somewhat misaligned with the marginal effect in (a, c), due to the correlation between site mean growing degree days and planting dates (see text). The marginal effects line up with the residuals of each in predictor residual plots (Fig. S7B,E). Degree days in this figure were calculated as cumulative growing degree days for the 70 d past planting (flowering-GDD); for analyses with degree days calculated over a fixed period (site-GDD), see Supporting Information Table S1 and Figs S10, S11.

Our model shows that temperature across sites and years as well as planting date have the biggest impact on the optimal flowering time of sunflower. To illustrate the impact of these three

factors on optimal flowering time, we developed a heat map (Fig. 6), which shows the model predictions of optimal flowering time (color) across sites in different climates (different average

Table 1 Effects of six climate predictors on the optimal flowering days past planting (DPP) for sunflower.

Model	Predictor	Slope	95% CI
DPP	GDD-site mean	−0.032	−0.051, −0.012
	GDD anomaly	−0.053	−0.078, −0.028
	Precip site mean	0.025	−0.037, 0.087
	Precip anomaly	−0.006	−0.03, 0.017
	First frost anomaly	0.029	−0.089, 0.144
	Planting date	−0.105	−0.237, 0.03

Bold text indicates that the slope is statistically clearly different from zero.

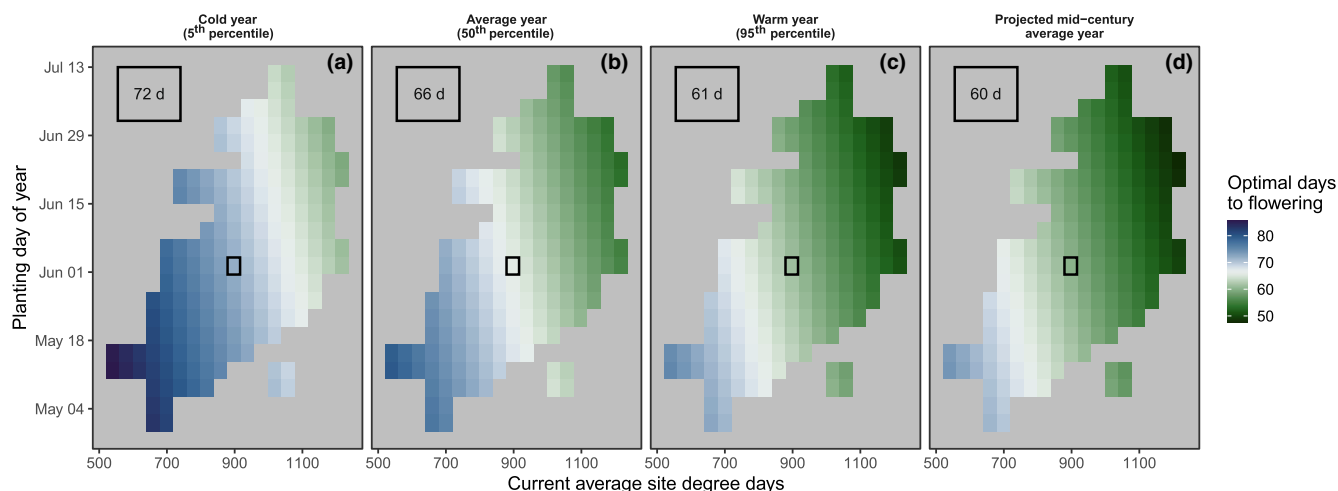


Fig. 6 Model predictions of optimal number of days to flowering across a range of planting days, average site degree days, and yearly anomalies. Predicted optimal days to flowering are indicated by the color of each cell. The black square outline on each panel shows a site with an average of c. 900 growing degree days planted on the same day in four scenarios: (a) a colder than average year, (b) an average year, (c) a warmer year, (d) and an average mid-century year (2040–2069) as projected from an intermediate emissions climate change model. The number in the top left of each panel shows the predicted optimal days to flowering for a site within the black outlined cell indicated on each panel.

growing degree day accumulations (x -axis)) and planting dates (y -axis), and in warm and cold years at each site (growing degree day anomalies (panels)). Hotter sites, hotter years, and later planting dates require genotypes that flower more quickly (Fig. 6a–c).

We found very similar results for models in which the dependent variable was the optimal flowering day of the year, rather than days after planting (Fig. S8; Table S1). In the model with optimal flowering day, the direction of the association with planting date switched, and the effect became stronger (Fig. S8C), such that for each day that planting is delayed, the optimal flowering date is delayed by > 1 d. The results of the two models are thus complementary: if planting is moved 5 d later, the optimal flowering date is predicted to be 4 d later, indicating that faster varieties are needed for later planting dates.

The results were similar and conclusions identical when sites were grouped by exact location, rather than county (Fig. S9; Table S2).

Warmer sites tended to have later planting dates than colder sites. This correlation held for both cumulative growing degree days for the 70 d past planting (flowering-GDD) and also for a fixed 125-d period (site-GDD) (Fig. S7A). Because planting later

in the spring means that plants are growing in a hotter part of the year, sunflowers at warmer sites are subjected to hotter temperatures, not only because it is a hotter site, but also because they are planted in a hotter part of the year (Fig. S7A). However, our results and conclusions were not sensitive to the period over which growing degree days were accumulated, for both days to flowering and flowering day of year (Figs S10, S11; Table S1).

How will global warming impact the optimal flowering time?

Across the sunflower-growing region, the climate is projected to warm by mid-century to 1.85 SD above the current average growing degree days, based on an intermediate emissions scenario (RCP 4.5). For comparison, current extreme years (95th percentile) are only 1.5 SD from the current average, making the near-future climate more extreme than current extreme years.

The optimal days to flowering for a location with a constant planting date are projected to be about a week earlier than the current average years (Figs 6d, S12). These results indicate that faster flowering genotypes and/or altered agronomic practices will be needed in the future to maintain optimal flowering time.

Discussion

The fitness consequences of phenological shifts are crucial for understanding how both natural and agricultural systems respond to climate change. Across 46 years and eight states, we find that the flowering time of sunflower is an important determinant of seed biomass production (yield), that the flowering time associated with the highest yield (optimal flowering time) is highly variable both among sites and between years within a single site, and optimal flowering occurs earlier in warmer conditions. As phenological shifts are the most prevalent response to climate change, the understanding of the climate drivers of optimal phenology is important for both wild plants and crop species facing increasingly warm temperatures and altered climate patterns. Our results suggest that both shifting flowering timing, along with altered agronomic practices in agricultural systems (specifically, earlier planting than current practice), are important in achieving optimal flowering time in future climates. We highlight some important implications of our results for both agricultural and natural systems.

Our analyses show that the timing of flowering is important for productivity, explaining an average of *c.* 9% of the variation in yield within a trial. Since there are many factors that contribute to yield, including many other plant traits, climate and soil conditions, pollination and herbivory, and agronomic factors, it is remarkable that we find that flowering time alone explains so much of yield. Additionally, even relatively small shifts in flowering time led to marked decreases in yield within trials. This indicates that breeding for optimal flowering timing will remain important for improving yield and yield stability. In many natural systems, phenology explains even more variation in fitness than we found here for cultivated sunflower (Austen & Weis, 2015; Zettlemoyer *et al.*, 2024). In natural systems, flowering may be spread over a longer period than in this commercial crop data, where the spread was often < 2 weeks. The sunflower genotypes in this dataset are commercially available sunflower varieties that are already considered 'elite' genotypes, and varieties included in a specific field trial are selected specifically because they are expected to perform well in that environment, based on previous trials. Thus, the proportion of yield explained by flowering time in these trials reflects prior knowledge about varieties that typically do well in certain locations, and is likely much less than might occur if a broader range of genotypes were planted in each location. This likely also explains why R^2 decreases over the 46 years of the study, as breeding has improved genotypes by removing suboptimal phenological variation and choosing varieties for broad adaptation across locations has improved. Nevertheless, our quantification in two separate analyses of the importance of flowering time in this species demonstrates that timing does explain substantial variation in performance. For natural species, phenology may play an even greater role in fitness than our data show for agricultural sunflower due to increased natural variation, a likely consequence of selection for niche environments rather than broad adaptation.

An emerging pattern from wildland systems is that natural selection tends to favor earlier timing of spring phenological

events (Ehrlén & Münzbergová, 2009; Anderson *et al.*, 2011; Munguía-Rosas *et al.*, 2011; Austen & Weis, 2015; Radchuk *et al.*, 2019), potentially due to climate warming shifting fitness landscapes (Anderson *et al.*, 2012). By contrast, in domesticated sunflowers, we did not find earlier flowering was consistently favored within trials, being favored in only 8% of trials. Later flowering was favored in twice as many trials as early flowering, and in the majority of trials (55%) in our dataset, the genotypes were suitable for the environment and flowered near the optimal time. There are several potential explanations for these results. First, the Great Plains have warmed less than many other regions over the past few decades (Bromley *et al.*, 2020), and it is possible that substantial shifts in flowering timing have not yet been necessary. Second, breeding of sunflower genotypes may have kept up with the relatively moderate rate of global change in this region. Third, sunflowers flower earlier under warm conditions (Aiken, 2005; Casadebaig *et al.*, 2011), and this plasticity may allow them to keep pace with warming. Last, basic physiology suggests that flowering cannot advance indefinitely without a fitness cost, particularly in semelparous annuals where flowering directly precedes senescence. Most of the studies of phenology in wild systems focus on iteroparous perennials, which can advance phenology without shortening lifespan, but in most annuals earlier flowering necessarily reduces lifespan, and the total time to photosynthesize resources for reproduction. Thus, while earlier flowering may be beneficial to avoid later-season stresses, the costs in terms of lost photosynthesis are substantial. The sunflower data here reflect that tradeoff, with the relationship between yield and flowering time being quadratic in the majority of cases. We propose that a quadratic relationship between yield or fitness and reproductive timing is likely common to most natural and agricultural systems, given enough data to detect these patterns.

While we found that earlier flowering phenology was not consistently favored, we did find a strong relationship between optimal flowering time and temperature, with warmer conditions leading to earlier optimal flowering time. Our results suggest that plasticity for flowering time in response to temperature, which leads to earlier flowering time under warm conditions in many natural systems (Stuble *et al.*, 2021) and domesticated sunflower (Aiken, 2005; Casadebaig *et al.*, 2011), is likely beneficial, since the optimal flowering time also shifts earlier in warmer sites and in warmer years. Earlier flowering timing in warming conditions can be achieved by selecting genotypes that consistently flower earlier or genotypes that shift earlier under warm conditions (plasticity, or genotype $\times$ environment interactions). Our analyses did not distinguish between these two processes, and we acknowledge that both mechanisms may play an important role and be important targets of selection.

We considered how precipitation and the timing of frost impact optimal flowering time, since genotypes that flower at a time that avoids these stressors may have higher fitness. However, these factors did not drive variation in optimal flowering time, though trials in years with early frost were more likely to have all genotypes flower later than optimal and lower yield. Sunflower is known to be relatively drought-tolerant crop (Hussain

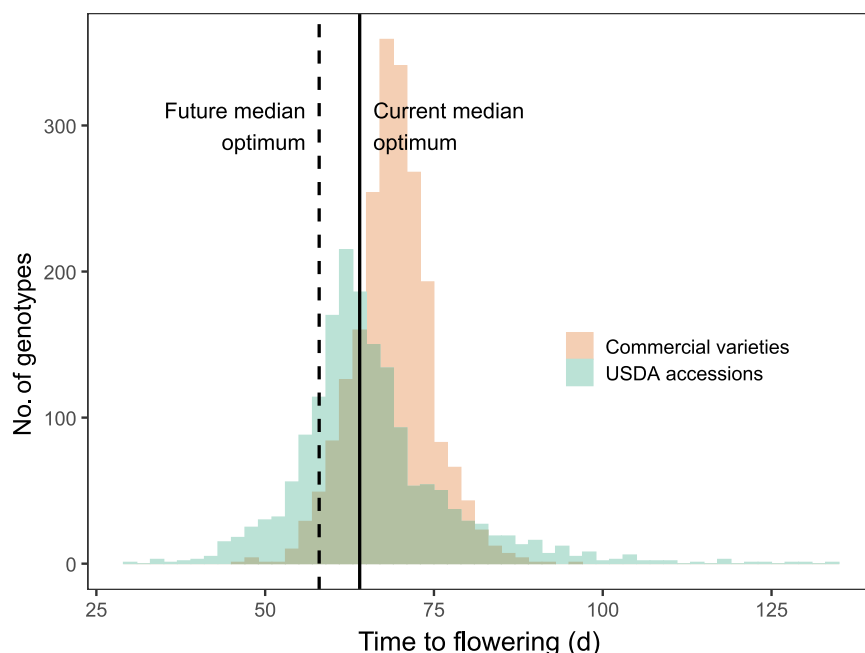


Fig. 7 Average days to flowering for each genotype of commercial sunflower varieties and USDA accessions. USDA accessions were evaluated in Ames, IA, from 1960 to 2014. The vertical solid line shows the current median optimal number of days to flowering, and the dashed line shows the median optimal days to flowering predicted for mid-century.

et al., 2018). In crops or wild plant species with greater drought sensitivity, precipitation (or irrigation) may play a stronger role in the fitness–phenology relationship than found here. Future studies contrasting the differences in optimal flowering time between dryland and irrigated agricultural systems and wild species with differing drought or frost tolerances may further illuminate additional environmental drivers of optimal flowering time.

About one-third of the variability in global crop productivity can be explained by variation in climate (Ray *et al.*, 2015), and thus, confronting climate change in agricultural systems is a major ongoing challenge (Lou *et al.*, 2024). The climate is predicted to warm the Great Plains region by *c.* 3°C by mid-21st century above historic averages (Kluck *et al.*, 2023), representing a shift in average conditions that exceed the 95th percentile years experienced over the past 40 years. While the climate warming hole in the Great Plains region may have reduced the impact of climate change in the 20th century, temperatures have already begun to rise. Modifying agronomic practices including crop phenology and planting dates has the potential to mitigate climate stressors (Tariq *et al.*, 2018; Krstić *et al.*, 2023). Our results imply that breeding for a climate-resilient future should focus on the development of varieties that can appropriately match hotter future climates. The median optimal days to flowering that we project under a future climate scenario is 58 d, but only 2% of commercial varieties in our dataset took an average of 58 d or less to flower (Fig. 7). However, there is more genetic variation available in genetic resource repositories, including the USDA National Plant Germplasm System (GRIN-Global, 2025), in which 16% of accessions took 58 d or less to flower. While it is difficult to precisely quantify the available genetic variation from the GRIN database and compare it with our dataset on the basis of growing degree days (due to unknown planting dates and climate conditions of the GRIN trials), this nonetheless represents an important source of genetic variation for ongoing breeding of

sunflowers with shorter flowering times (e.g. Hulke & May, 2018).

In addition to breeding for faster average flowering times, breeding for phenotypic plasticity of flowering time in response to temperature is also likely to be an important component of achieving optimal flowering time in hotter climates, especially across several years and in a landscape predicted to have more variability in the future. Many sunflower genotypes already flower faster in warmer conditions, but it is not yet known whether breeding for flowering time plasticity will improve yield stability. However, our data suggest that plasticity alone is not enough to achieve optimal flowering time, since in many trials in our dataset, all genotypes still flowered too late. Understanding the drivers and effects of plasticity on flowering time, and their consequences on yield and yield stability, will be crucial for informing future plant breeding (Cleland *et al.*, 2012; Ramirez-Parada *et al.*, 2024).

Another way to maintain optimal flowering time in warmer climates is to shift planting dates earlier. In our dataset and in regular commercial growing operations across the United States, planting is often *later* in hotter sites than at cooler sites, and early planting is not currently used at warmer sites (bottom-left grayed-out area of Figs 6, S13), which compounds the heat stress experienced by plants in hotter environments. However, planting earlier in warmer locations could potentially reduce heat stress during flowering time and reduce pest problems (Oseto *et al.*, 1987; Krstić *et al.*, 2023), while still allowing favorable temperatures. Earlier planting dates would also allow later-flowering genotypes to be planted, which are more common commercially (Fig. 7). There are many factors that control planting dates that will not make this feasible in all cases, like the timing of harvest of winter crops, waiting for fields to be dry enough to plant, and the availability of equipment to plant seeds. While sunflower planting occurs later in warmer climates in the United

States, planting dates are earlier in warmer climates in Europe (Sacks *et al.*, 2010). Shifts in agronomic practices, including planting date, will be necessary in the future. Although our study focused on the productivity of sunflower alone, we recognize that decisions about planting dates and variety selection must take into account the economic value of the entire system, which in some cases may mean selecting a combination of planting dates and flowering times that allows double-cropping or rotation needs (Gesch *et al.*, 2022).

Multi-environment agricultural trials represent an underutilized resource of large datasets for examining the relationships between phenology, performance, and the environment. High levels of interannual variation may mask the effects of long-term trends (i.e. climate change) in small datasets, so decades-long replicated experiments such as these have the potential to uncover patterns that would not be clear from short-term manipulations. These datasets can complement knowledge in natural systems, where large long-term experiments can be difficult to perform and fund. Even with large datasets from multi-environment trials, the optimum phenotype can be difficult to locate precisely within a single trial, particularly when most individuals are on the same side of the optimum, leading to large uncertainties in the exact optimum for any single environment (Chevin *et al.*, 2013). Our analytical approach – combining the measurement error and censoring models – allowed us to leverage a large number of trials grown in diverse environments, even when we could not precisely estimate the optimal flowering time in individual trials.

Here, we focus on a single annual species, sunflower, which predominantly outcrosses in the wild and is commercially bred to be planted as F1 hybrids. Many other crops, such as perennials (e.g. apples), may have very different relationships between optimal phenology and climate. Our work is generally in agreement with other studies of annual crops, which show that optimal phenology depends on climate. For example, optimal flowering and sowing periods of wheat in Australia depend on seasonal water supply and temperature extremes (Flohr *et al.*, 2017). In canola and barley, optimal flowering time was driven by rainfall and temperature (Lilley *et al.*, 2019; Liu *et al.*, 2020).

Domesticated and wild sunflower are the same species, *Helianthus annuus*, and occur over the same geographic range. As a result, we hypothesize that many of the same selective forces are likely to apply in wild populations of this and other species in this region. However, there are some key differences between wild and crop plant phenology that make some comparisons between the two difficult. First, the timing of planting is controlled in agricultural systems and can be manipulated with respect to local climate and seasonal forecasts, whereas spring emergence timing in wild species is more variable and may also be under selection. Additionally, many crop species are planted in monoculture with little to no plant interspecific interactions, but these interactions are common in natural systems and may influence the optimal timing of flowering in wild species. Acknowledging these caveats, our results have important implications for natural, as well as agricultural species, in terms of the direction and magnitude of phenological shifts necessary to maintain fitness under a warming

climate. We encourage more studies in natural systems to explore the role of optimal timing in wildland species, particularly in annual species, as well as direct comparisons of wild and agricultural conspecifics grown in the same regions to better understand how agricultural environments compare to nearby wild spaces.

In situ data are just one of many approaches to understand the consequences of changing phenology for fitness and yield. For example, many studies have used process-based crop models, which estimate optimal crop phenology and yield based on solar radiation, water, and temperature during the growing season (Flohr *et al.*, 2017; Lilley *et al.*, 2019; Liu *et al.*, 2020). Another approach, used for tree species, relies on satellite imagery to parameterize models of the causal relationship between climate, phenology, and fitness (Gauzere *et al.*, 2020). These types of models have made important advances in understanding the relationships between phenology and fitness, but are typically calibrated with experimental data from relatively few locations and genotypes and rely heavily on simulations, which necessarily make some assumptions. Analyzing long-term performance field trial data, such as we present here, offers a complementary approach to understand the impact of climate on optimal flowering time across both a large number of sites and years. Our approach could be done with other crops that undergo field performance trials. We also emphasize the need for multi-environment performance trials to consistently record phenology and agronomic practices like planting date. This information would make multi-environment trial data more accessible for synthesis approaches such as we present here.

Climate change already has substantial impacts on both natural and agricultural systems. The timing and plasticity of reproduction will be an important determinant of whether wild populations will be able to persist, and whether agricultural systems will be reliably productive. Further work remains to develop climate-resilient agricultural systems and also to illuminate the climate drivers of optimal flowering time in natural systems across varying environmental conditions.

Acknowledgements

The authors gratefully acknowledge Alecia Pardue, Leanne Matthesen, Mike Ostlie, Bryan Hansen, Jane Lingenfelter, Shana Forster, John Rickertsen, James Staricka, and the many scientists, agronomists, technicians, student trainees, and extension experts who have added to the collection of data over the last 5 decades, in order to advance scientific progress on farms and ranches. Mention of trade names or commercial products in this article is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the U.S. Department of Agriculture. USDA is an equal opportunity provider and employer. This work was supported by NSF grant 2308194 and USDA-Agricultural Research Service CRIS project 3060-21000-047.

Competing interests

None declared.

Author contributions

EIC was responsible for conceptualization, formal analysis, visualization and writing – original draft preparation. MAM contributed to data curation, software, visualization and writing – review & editing. CKK contributed to conceptualization, funding acquisition and writing – review & editing. BSH was involved in conceptualization, funding acquisition, data curation and writing – review & editing. NCK contributed to conceptualization, funding acquisition, resources and writing – review & editing. SCE was responsible for conceptualization, funding acquisition, data curation, formal analysis and writing – review & editing.

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Data availability

All sunflower data and code that support the findings of this study are available on Zenodo at doi: [10.5281/zenodo.17547241](https://doi.org/10.5281/zenodo.17547241). The climate datasets used in this study are available from Daymet at doi: [10.3334/ORNDAAC/2129](https://doi.org/10.3334/ORNDAAC/2129). Additional data on the phenology of sunflower accessions used in the analyses are available from GRIN-Global at <https://npgsweb.ars-grin.gov/gringlobal/descriptordetail?id=79038>.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Climate trends.

Fig. S2 R^2 and sample size association.

Fig. S3 Example Gaussian and linear models of optimal flowering time and yield.

Fig. S4 R^2 across years.

Fig. S5 Probability of all genotypes flowering timing being suitable, early, or late across years.

Fig. S6 Distribution of optimal flowering times (days past planting).

Fig. S7 Planting day-climate correlations and trends of optimal flowering with residuals of GDD and planting day.

Fig. S8 Association of optimal flowering time and climate with flowering day of year.

Fig. S9 Association of optimal flowering time and climate, in days past planting, grouped by exact location.

Fig. S10 Association of optimal flowering time with climate, in days past planting, with GDD-site.

Fig. S11 Association of optimal flowering time with climate, with day of year and GDD-site.

Fig. S12 Heatmap of predictions of optimal flowering day of year across planting day of year and current site-GDD.

Fig. S13 Average planting dates for sunflower across the US growing region.

Methods S1 Description of measurement error-censored modeling technique and predictor residual methods.

Table S1 Results of climate-optimal flowering time association models with grouping by county.

Table S2 Results of climate-optimal flowering time association models with exact location grouping.

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